

KUSHAL ESHWAR REDDY GUDIBANDI

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PROFESSIONAL SUMMARY

Software Development Engineer with experience in building scalable, high-performance systems using Java, React, and Python. Strong foundation in data structures, algorithms, and distributed systems. Proficient in developing microservices, secure APIs, and cloud-native applications on AWS. Familiar with mobile development using React Native and actively learning Go for backend scalability. Passionate about improving user experiences and supporting live systems in production environments.

TECHNICAL SKILLS

Languages	Java 8, Java 11, Java 17 (JDK 1.4/1.5/1.6/1.7/1.8), Python, Node.js, JavaScript, TypeScript, Go, C++, Shell Script
Methodologies/Design patterns	Singleton, Agile, Scrum, Test Driven Development
Web Technologies	Hibernate, Struts, Spring, JDBC, HTML5, DHTML and XML, PHP, CSS3, Web Services, WSDL, SOAP, Kafka, Spring Boot, Express.js
Cloud Platforms:	Google Cloud Platform (Compute Engine, Cloud SQL, Pub/Sub), Serverless Framework
Frameworks & Libraries:	Spring Boot, Spring MVC, Hibernate, React JS, Angular, Flask, Django, Express.js, Bootstrap, Redux, Spring JMS, WebSockets
Application/Web Servers	JBoss, WebLogic, Web Sphere, Apache Tomcat Server
Web & API Technologies:	GraphQL, JSON, AJAX, WebSockets, Kafka Streams, SOAP, RESTful Web Services
IDE Tools	VS Code, IntelliJ, Eclipse, PyCharm
ORM Mapping	Hibernate, IBATIS
Databases	MongoDB, MySQL, PostgreSQL, DynamoDB, Oracle, RDS, Redis, NoSQL & Relational Data Modeling, Query Optimization, Caching, Sharding
Reporting/Messaging Tools	Kafka, RabbitMQ, ActiveMQ
Cloud	AWS (Lambda, EC2, S3, DynamoDB, API Gateway, Amazon ECS), Azure, AWS IAM Policies, AWS Lambda, Azure Pipelines, Azure Repos
DevOps & CI/CD Tools:	Helm, GitHub Actions, CloudWatch, Stackdriver
Version Control Tools	CVS, SVN, Clear Case, Git

Testing	Selenium, Karma
Operating System	Windows, Linux, Unix, Mac, Solaris
Infrastructure & DevOps Tools:	Kubernetes, Docker, Jenkins, CI/CD Pipelines, Git, Azure DevOps, Terraform, YAML, AWS CodePipeline, Maven, Gradle, Ansible
Monitoring & Logging:	Prometheus, Grafana, Splunk, SonarQube
ML Frameworks:	TensorFlow, PyTorch, Scikit-learn, Keras, Struts
Machine Learning & Analytics:	Pandas, NumPy, Apache Spark, Data Pipelines, Anomaly Detection, Predictive Analytics
Testing Tools	JUnit, TestNG, Selenium, Cucumber, Karma, PyTest, Load Testing (JMeter), Unit Testing, Integration Testing, Regression Testing
Test Automation	Selenium WebDriver, Postman, REST Assured, Automation Frameworks (TDD/BDD), Continuous Testing with Jenkins, GitHub Actions, and CI/CD Pipelines
Monitoring & Logging:	ELK Stack, CloudWatch, Stackdriver
Accessibility & Performance Optimization:	WCAG & ARIA Compliance, SEO Optimization, Lazy Loading, Code Splitting, Caching, Load Balancing, Responsive UI
Core Computer Science:	Data Structures (Trees, Graphs, Heaps, Linked Lists), Algorithms (Sorting, Dynamic Programming, Search), System Design, Object-Oriented Programming (OOP), Multithreading, Concurrency Control, Memory Management
Additional Design Patterns / Methodologies:	Factory, Strategy, MVC, Microservices Architecture, Event-Driven Design

Certifications

- AWS Certified Developer.
- AWS Certified Solutions Architect – Associate.

PROFESSIONAL EXPERIENCE

Software Engineer

Department of Revenue, Florida

Nov 2024 – Present

Project Name: IBM Watson Outreach Automation

Project Description: The **IBM Watson Outreach Automation project** aimed to develop a microservices architecture for automated, evidence-based healthcare communications. By leveraging IBM Watson's

advanced AI capabilities, the system was designed to deliver tailored health information and reminders to patients, thereby enhancing engagement and adherence to treatment plans. The architecture seamlessly integrated components such as patient data management and communication scheduling, utilizing **Spring Boot** to build the microservices and **MongoDB** for managing patient data. Each microservice was crafted for specific tasks, like generating personalized messages and tracking responses, facilitated by **RabbitMQ** for reliable messaging between services. Additionally, the project employed **Docker** for containerization and **Kubernetes** for orchestration, enabling rapid deployment of new features and ensuring scalability. This innovative solution significantly improved the effectiveness of healthcare outreach programs.

Responsibilities:

- Developed and deployed **15+ microservices** using **Spring Boot**, handling over **1 million API requests per day**, applying principles of **data structures, algorithms, and load balancing** to maintain low latency under heavy traffic. Improving system scalability Built backend Go services (PoC) to evaluate concurrency strategies using goroutines and channels, enhancing parallel request processing efficiency by 20%.
- Built **fault-tolerant microservices** and **real-time streaming pipelines** using Kafka and Spark, enabling efficient data flow between subsystems.
- Applied **algorithmic optimizations** (hash-based indexing, caching, and parallelism) that reduced API response times by 30%.
- Integrated **AI/ML models (IBM Watson, Python scikit-learn)** for message classification and predictive analytics, improving targeting accuracy by 25%.
- Developed **cross-platform accessible features** following WCAG 2.1 standards to ensure usability for all users.
- Deployed systems in **Unix/Linux** environments using Kubernetes and Docker, ensuring reproducible builds and container security.
- Designed **20+ Spring Rest Controllers**, reducing API response times by **30%** and improving data access performance using **JPA & Hibernate ORM**.
- Created **10+ dynamic forms** in **Angular**, improving data handling efficiency by **25%** through advanced form validations.
- Built **15+ responsive web pages** using **Angular** and **Bootstrap**, increasing mobile traffic by **35%** and improving page load times by **30%**.
- Contributed to adoption of a micro-frontend architecture across 5 teams, allowing for independent deployments of application modules and enhancing team scalability, reducing time to market by 20%.
- Developed **20+ web pages** using **React JS**, resulting in **30% faster page load times** and a **25% increase** in user engagement.
- Designed and implemented a **global state management** solution using **React's Context API**, optimizing data flow and reducing prop drilling, which improved application maintainability and reduced state management-related bugs by 15%.
- Achieved **95% browser compatibility**, reducing device-specific bugs by **50%** and increasing user satisfaction by **15%**.
- Integrated **Spring Boot** with Spring Data to create data access layers that supported both **SQL** and **NoSQL** databases, resulting in a 25% decrease in data retrieval times and improved application responsiveness.

- Utilized Spring Security to implement **OAuth 2.0** and **JWT-based authentication**, securing APIs and reducing unauthorized access attempts by 50%, thereby enhancing application security and compliance.
- Acted as a liaison between frontend and backend teams, ensuring smooth integration of **Node.js APIs** with **React** and **Angular** applications, reducing integration issues by **25%**.
- Built and maintained 5+ mobile-responsive features using React Native, improving usability across iOS and Android platforms.
- Collaborated with cross-functional teams to design scalable systems used by 1M+ users daily, ensuring operational excellence and fast iteration cycles.
- Designed and implemented event-driven architectures using **Node.js** and **Kafka**, handling over 2 million messages per day with minimal lag.
- Automated **ETL** workflows using **Node.js** and **Apache Spark** for processing large datasets, reducing data pipeline execution time by **50%**.
- Implemented modular **microservices** using **Node.js** and **Express.js**, enabling seamless integration across services and reducing code duplication by **35%**.
- Configured and optimized **MongoDB** and **MySQL** connections for **Node.js** applications, resulting in a **40%** reduction in database query times.
- Integrated **10+ Node.js/Python applications** with **MongoDB** and **MySQL**, improving query response times by **40%**.
- Coordinated with front-end developers to integrate **server-side logic** with **15+ components**, improving development speed by **25%**.
- Increased test coverage to **95%** using **TDD**, reducing post-release defects by **30%**.
- Configured **Kafka** clusters to process over **2 million messages per day**, reducing message lag time by **35%** and achieving **99.9% uptime**.
- Developed **30+ reusable React components**, improving development efficiency by **40%** and reducing redundant code.
- Developed a microservices architecture using **Spring Cloud**, enabling service discovery with **Eureka** and centralized configuration with **Spring Cloud Config**, which streamlined deployment processes and improved system reliability by 30%.
- Implemented **CI/CD pipelines** in **Jenkins** with **Selenium**, **Cucumber**, and **SonarQube**, reducing deployment time by **30%** and improving code quality by **25%**.
- Reduced deployment failures by **20%** and accelerated release cycles by **35%** through **Azure CI/CD pipelines**.
- Improved version control management by **30%** using **Azure Repos**, reducing code merge conflicts by **25%**.
- Built backend Go services (PoC) to evaluate concurrency strategies using goroutines and channels, enhancing parallel request processing efficiency by 20%.
- Achieved **95% unit test coverage**, reducing production bugs by **30%** with thorough unit testing using **JUnit** and **Mockito**.

Environment: Java 11, Python, Node.js, Express.js, Flask, Django, JavaScript, React JS, Angular, YAML, Terraform, AWS (Lambda, EC2, S3, DynamoDB, API Gateway, Amazon ECS), Azure (Azure Resource Groups, Azure Management APIs, RBAC), Kubernetes, Docker, Jenkins, CI/CD Pipelines, Git, Prometheus, Grafana, ELK Stack, MongoDB, MySQL, PostgreSQL, Spring Boot, Microservices Architecture, AWS Lambda, Git, Azure Repos, Role-Based Access Control (RBAC), AWS IAM Policies, HTML5, CSS3, JavaScript, Redux, TDD, JUnit, Selenium, Cucumber, RESTful APIs, GraphQL.

Software Developer

Aramark, Florida

Sep 2022 – Apr 2024

Project Name: Academic Enterprise Modernization Platform

Project Description:

The Academic Enterprise Modernization Platform initiative focused on modernizing legacy academic and administrative systems by transitioning from monolithic C++ and Java applications to a modular, service-oriented architecture. The project aimed to improve system scalability, reliability, and user experience across student and enterprise portals. It introduced modern front-end frameworks such as React and Redux for responsive interfaces and enhanced CI/CD pipelines for faster and more reliable deployments. The system also integrated analytics-driven monitoring to ensure performance consistency and high availability under heavy usage.

Responsibilities:

- Developed and debugged backend modules using **C++ and Java**, improving application response times by **25%** through multithreading optimization and efficient memory management.
- Refactored legacy academic software into modular microservices, leveraging **object-oriented design patterns** (Singleton, Factory, Strategy) for extensibility.
- Implemented **search and data retrieval algorithms** to enhance indexing speed and query accuracy for enterprise reporting.
- Developed **unit and integration testing frameworks** to validate algorithm correctness and concurrency safety across distributed modules.
- Applied **networking fundamentals** to improve client-server communication reliability over multi-region setups.
- Deployed **automated build and test pipelines** ensuring consistent integration, regression, and performance validation.
- Designed and implemented **React.js** front-end components with **Redux** state management, achieving **98% cross-browser compatibility** and enhancing UI consistency.
- Applied **Test-Driven Development (TDD)** using **JUnit, Mockito, and Selenium**, improving code coverage by **35%** and reducing production defects by **40%**.
- Enhanced CI/CD pipelines using **Jenkins** and **GitLab CI**, automating build, test, and deployment workflows for a **25% faster release cycle**.
- Collaborated with DB engineers to optimize **PostgreSQL** and **MySQL** queries, reducing transaction latency by **20%** for high-traffic workloads.
- Containerized microservices using **Docker**, standardizing build environments and reducing setup issues by **50%**.
- Integrated **Grafana, Prometheus, and Splunk** dashboards for proactive system health monitoring and root cause analysis.
- Partnered with product and UX teams to implement accessibility and usability improvements, leading to higher user adoption and satisfaction.
- Mentored junior developers in debugging, OOP best practices, and agile collaboration, enhancing team productivity and code quality.
- Authored detailed technical documentation and SOPs to standardize knowledge transfer across development and QA teams.

Environment:

C++, Java 8/11, React JS, Redux, Bootstrap, Spring Boot, Spring MVC, RESTful APIs, MySQL, PostgreSQL, MongoDB, Docker, Jenkins, GitLab CI, Maven, Gradle, Selenium, JUnit, Mockito, TDD, Grafana, Prometheus, Splunk, Agile/Scrum.

Software Developer

United cloud technologies, Dallas

Jun 2024- Oct 2024

Project Name: Next Generation Client Solutions

Project Description: The **Next Generation Client Solutions** project aimed to transform client applications by implementing a robust microservices architecture that utilized **Spring Boot** for developing modular services and **AWS** for cloud infrastructure. By decomposing monolithic systems into independent microservices, the project enhanced scalability and performance, allowing for more efficient development and deployment. AWS services ensured high availability and optimized operational costs, while **Docker** facilitated containerization for streamlined deployments. Automated **CI/CD pipelines** were integrated to enable rapid updates and faster time-to-market for new features. This initiative significantly improved user experiences, reduced response times, and enhanced overall reliability, positioning clients for success in a dynamic digital landscape.

Responsibilities:

- **Configured and optimized Angular CLI**, reducing project setup time by **30%** and enabling more efficient development workflows.
- Built backend proof-of-concept services in Go for experimentation with concurrent processing and performance tuning.
- Engineered complex, nested component hierarchies with advanced **input/output decorators** and **dynamic component loading strategies**, enhancing **modularity** and reuse across large-scale applications, resulting in a 30% reduction in development time for new features.
- Designed and implemented **scalable REST and GraphQL APIs** supporting distributed microservice communication across AWS regions.
- Developed efficient **C++/Java** services for low-latency data access and optimized concurrent workloads using threads and async I/O.
- Built **ETL and data-streaming solutions** integrating Spark Structured Streaming, Kafka, and AWS Glue for multi-source data ingestion.
- Participated in **code reviews** emphasizing clean architecture, algorithmic efficiency, and security compliance.
- Contributed to **information retrieval and search optimization** workflows, improving query ranking and response time for large datasets.
- Explored **machine-learning-driven anomaly detection** models to monitor microservice health metrics.
- Mastered **RxJS** for **reactive programming**, implementing observables and operators to manage asynchronous data streams, which improved application responsiveness and reduced load times by 25% in real-time data scenarios.
- Focused on creating clean, robust code for **React JS applications**, achieving **95% cross-browser**

- **compatibility** and reducing reported bugs by **40%** across devices.
- Assessed and implemented changes to core UIs and middleware services, leading to **25% faster response times** and improved application stability.
- **Created Terraform modules** to provision and manage cloud resources, reducing manual configurations by **30%** and improving deployment speed by **25%**.
- Integrated AWS services like S3, API Gateway, and DynamoDB with **Node.js** applications to build cloud-native solutions, reducing operational overhead by **25%**.
- Developed server-side applications using **Node.js** and **Python**, processing **1 million+ API requests per day**, ensuring **99.9% uptime**.
- Implemented **RESTful APIs and microservices** using **Express.js (Node.js)** and **Flask/Django (Python)**, reducing API response times by **20%** and improving scalability.
- Developed enterprise-level applications using the **Spring Framework**, implementing dependency injection and **aspect-oriented programming (AOP)** to enhance modularity and maintainability, which reduced development time by **30%**.
- Leveraged **Spring MVC** to build RESTful web services that handle concurrent requests, achieving a response time improvement of **40%** and enhancing overall application performance.
- Developed and maintained **Kubernetes manifests (YAML files)**, improving Java application deployment efficiency by **25%** and ensuring **99.9% uptime**.
- **Optimized pod scaling** and resource management for Java microservices in **Kubernetes**, reducing cloud costs by **20%**.
- **Deployed and managed microservices** on **AWS** using **Docker** and orchestration tools like **Amazon ECS** and **Kubernetes**, increasing scalability by **30%**.
- Designed and deployed **AWS Lambda functions**, reducing server costs by **20%** and improving processing time by **15%**.
- Integrated **Lambda functions** with **API Gateway, S3, and DynamoDB**, creating a **serverless architecture** that reduced operational overhead by **25%**.
- Enabled automation for platform-specific applications using **Terraform**, leading to a **20% reduction** in manual intervention and a **30% improvement** in deployment efficiency.
- Established **Azure Resource Groups** and implemented **role-based access control (RBAC)**, improving security compliance by **35%** and reducing unauthorized access incidents by **40%**.
- Developed and tested scalable **microservices**; implemented unit and integration tests using **JUnit and Mockito**.

Environment: Java8/11, JavaScript, TypeScript, Node.js, Spring Boot, Spring MVC, Hibernate, React JS, Angular, Bootstrap, Redux, AWS (EC2, Lambda, S3, RDS, DynamoDB), Azure (VMs, Azure Pipelines, Azure Repos), Spring Boot, RESTful APIs, Docker, Kubernetes, AWS Lambda, Kafka, RabbitMQ, ActiveMQ, Git, Azure Repos, Jenkins, Azure DevOps, AWS CodePipeline, JUnit, Mockito, Selenium, Cucumber, SonarQube, MySQL, MongoDB, PostgreSQL, Docker, Kubernetes, Amazon ECS, Splunk, Prometheus, Grafana, ELK Stack, Maven, Gradle, Terraform, Ansible, TDD practices.

EDUCATION

Master of Science (MS): Computer Science: Florida State University, Florida

Bachelor of technology in computer science & engineering: SRM University, India

